

Key (attended stream)

VLM
Prefix

4D Future
Expert

World Knowledge
Expert

Action
Expert

Query
stream

VLM
Prefix

4D Future
Expert

World Knowledge
Expert

Action
Expert

Causal
self

Masked

Masked

Masked

Full
cross

Causal
self

Causal
UBCA

Masked

Full
cross

Causal
UBCA

Causal
self

Masked

Full
cross

Full
cross

Full
cross

Causal
self



Causal self-attention



Full cross-attention

Causal UBCA

Masked